

MatBERT. We finetune MatBERT on our benchmark dataset using its original tokenizer. We initially preprocessed the input text descriptions of MatBERT as described in 3.2. However, these preprocessing steps led to worse performance for MatBERT. The possible reason might be that since MatBERT can process only 512 tokens as input, the compressed information in 512 tokens that the preprocessing step gives, does not provide enough context for MatBERT. We therefore finetuned MatBERT with a batch size of 64 crystal descriptions, a learning rate of $5e-5$, and a dropout rate of 0.5 for 200 epochs using the Adam optimizer with a onecycle learning rate scheduler (Smith and Topin, 2019) for all the experiments.

4.2 LLM-Prop

For LLM-Prop, we finetune the original T5 encoder directly on our benchmark dataset without further pretraining it on domain-specific data. We first train the original T5 tokenizer on the benchmark data with the vocabulary size of 32k, then preprocess the data as described in 3.2. For regression tasks, we finetune LLM-Prop on normalized property values and denormalize the predicted values to calculate the error on the original values. Unless otherwise mentioned, we use z-score normalization when finetuning both LLM-Prop and MatBERT. Unless otherwise mentioned, we finetune LLM-Prop on 888 input tokens, using a batch size of 64, a learning rate of $1e-3$, and a dropout rate of 0.2 for 200 epochs using the Adam optimizer with a onecycle learning rate scheduler. For all models, we save the checkpoint of each epoch and evaluate on the test set with the checkpoint that gives the best performance on the validation set.

For regression tasks, we train both MatBERT and LLM-Prop with the mean absolute error (MAE) loss and evaluate them in terms of MAE error while the GNN-based models are trained with root mean square error (RMSE) loss and evaluated with the MAE error. For classification, we train them with binary cross entropy (BCE) loss and evaluate with the area under the ROC curve (AUC).

5 Discussion

5.1 Main results

LLM-Prop vs GNN-based models. We find that LLM-Prop outperforms all GNN-based baselines on both regression and classification tasks. For band gap prediction, LLM-Prop outperforms the best-performing baseline (ALIGNN) on both validation and test sets by approximately 8% and 4% improvements, respectively (see Table 2). For volume prediction, LLM-Prop also improves over the best-performing baseline (ALIGNN) on both validation and test sets by approximately 67% and 66%, respectively (see Table 3). The possible reason for this improvement might be that LLM-Prop can easily access the most important information for volume prediction, e.g. space group information, from the text descriptions compared to GNNs. This insight encourages two things: (1) the development of more text-based methods for crystal property prediction and the creation of high-quality text datasets needed for crystalline solids and (2) the development of new graph-based architectures that can incorporate more information, such as space group and Wyckoff sites. For